

Lab07VitorAugustoCosta267117

```
library(DBI)
library(RSQLite)

# Conectar a base de dados
conn <- dbConnect(RSQLite::SQLite(), "database.sqlite3")
```

```
query_quantos_professores <- "
  SELECT COUNT(DISTINCT i.id) AS quantidade_professores
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections s ON t.section_uuid = s.uuid
  JOIN course_offerings co ON s.course_offering_uuid = co.uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects sub ON sm.subject_code = sub.code
  WHERE sub.abbreviation = 'STAT';
"
cat("Quantidade de professores:\n")
```

Quantidade de professores:

```
print(dbGetQuery(conn, query_quantos_professores))
```

	quantidade_professores
1	379

```
query_quem_sao <- "
  SELECT DISTINCT i.name AS nome_professor
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections s ON t.section_uuid = s.uuid
```

```

JOIN course_offerings co ON s.course_offering_uuid = co.uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects sub ON sm.subject_code = sub.code
WHERE sub.abbreviation = 'STAT'
ORDER BY i.name;
"
cat("\nLista de professores:\n")

```

Lista de professores:

```
print(head(dbGetQuery(conn, query_quem_sao), 10))
```

```

      nome_professor
1      ABIGAIL BENZINE
2 ADIN-CRISTIAN ANDREI
3      AKICHIKA OZEKI
4      ALAN HUANG
5      ALBERTO DEL PIA
6      ALBRECHT KLEMM
7 ALEXANDER COVINGTON
8      ALEXANDER FISH
9      ALEXANDER KISELEV
10     ALYSSA DIGILIO

```

```

sql_gpa_cte <- "
WITH NotasEstatistica AS (
  SELECT
    i.name AS professor,
    c.name AS disciplina,
    (gd.a_count * 4.0 + gd.ab_count * 3.5 + gd.b_count * 3.0 +
     gd.bc_count * 2.5 + gd.c_count * 2.0 + gd.d_count * 1.0 + gd.f_count * 0.0) AS pontos,
    (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count +
     gd.c_count + gd.d_count + gd.f_count) AS alunos_avaliados
  FROM grade_distributions gd
  JOIN course_offerings co ON gd.course_offering_uuid = co.uuid
  JOIN courses c ON co.course_uuid = c.uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects sub ON sm.subject_code = sub.code
  JOIN sections s ON co.uuid = s.course_offering_uuid AND gd.section_number = s.number

```

```

        JOIN teachings t ON s.uuid = t.section_uuid
        JOIN instructors i ON t.instructor_id = i.id
        WHERE sub.abbreviation = 'STAT'
    )
"

# Professor mais difícil
query_prof_dificil <- paste0(sql_gpa_cte, "
    SELECT professor, SUM(pontos_totais) / SUM(alunos_avalidados) AS gpa_medio
    FROM NotasEstatistica
    GROUP BY professor
    HAVING SUM(alunos_avalidados) > 0
    ORDER BY gpa_medio ASC
    LIMIT 1;
")
cat("\nProfessor mais difícil:\n")

```

Professor mais difícil:

```
print(dbGetQuery(conn, query_prof_dificil))
```

```

      professor gpa_medio
1 JAMES D KUELBS    2.6375

```

```

# Professor mais fácil
query_prof_facil <- paste0(sql_gpa_cte, "
    SELECT professor, SUM(pontos_totais) / SUM(alunos_avalidados) AS gpa_medio
    FROM NotasEstatistica
    GROUP BY professor
    HAVING SUM(alunos_avalidados) > 0
    ORDER BY gpa_medio DESC
    LIMIT 1;
")
cat("\nProfessor mais fácil:\n")

```

Professor mais fácil:

```
print(dbGetQuery(conn, query_prof_facil))
```

```
professor gpa_medio
1 YAJUAN SI      4
```

```
# Disciplina mais difícil
query_disc_dificil <- paste0(sql_gpa_cte, "
  SELECT disciplina, SUM(pontos_totais) / SUM(alunos_avalidados) AS gpa_medio
  FROM NotasEstatistica
  GROUP BY disciplina
  HAVING SUM(alunos_avalidados) > 0
  ORDER BY gpa_medio ASC
  LIMIT 1;
")
cat("\nDisciplina mais difícil:\n")
```

Disciplina mais difícil:

```
print(dbGetQuery(conn, query_disc_dificil))
```

```
disciplina gpa_medio
1 Introduction to the Theory of Probability 2.886374
```

```
# Disciplina mais fácil
query_disc_facil <- paste0(sql_gpa_cte, "
  SELECT disciplina, SUM(pontos_totais) / SUM(alunos_avalidados) AS gpa_medio
  FROM NotasEstatistica
  GROUP BY disciplina
  HAVING SUM(alunos_avalidados) > 0
  ORDER BY gpa_medio DESC
  LIMIT 1;
")
cat("\nDisciplina mais fácil:\n")
```

Disciplina mais fácil:

```
print(dbGetQuery(conn, query_disc_facil))
```

	disciplina	gpa_medio
1	Sample Survey Theory and Method	4

```
# Encerrar a ligacao ao banco de dados  
dbDisconnect(conn)
```

horario

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-23 22:59:27 -03"
```